

Aprenentatge Automàtic per a Xarxes (ML4Net)

Seminar 2 - Support Vector Machine

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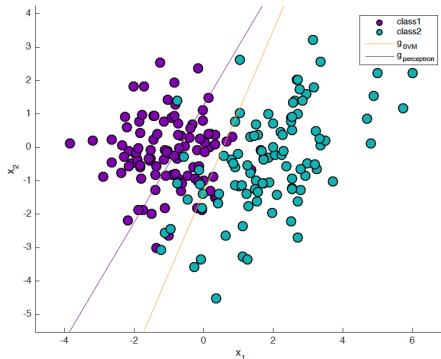


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Classification

Key concepts:

- **Features (x):** Input information for the predictions.
- **Labels (y):** Variables to be predicted (a discrete class).
- **Model (h):** Function $h(x) = y$ (set of weights and parameters) that maps x to y .
- **Goal:** Find the hyperplane that minimizes the classification error.



Support Vector Machine (SVM)

- We want to maximize the minimum distance between classes
- We do it optimizing the following:

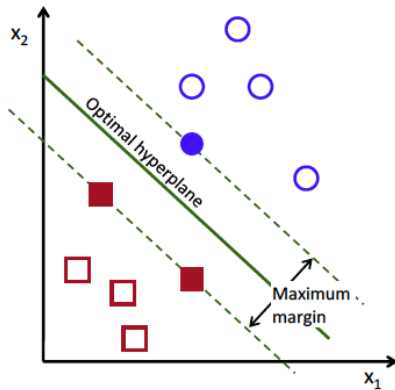
$$\omega^T x + b,$$

where $\omega \in \mathbb{R}^d$ is a weight vector and $b \in \mathbb{R}$ is a bias term

- How?

$$\text{Minimize } \frac{1}{2} \|\omega\|^2,$$

$$\text{s.t., } y_i(\omega^T x_i + b) \geq 1, \forall i = 1, \dots, n$$



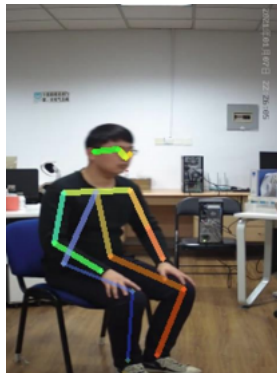
Cherkassky and Mulier, 1998

Dataset (I)

How were the data generated?^a

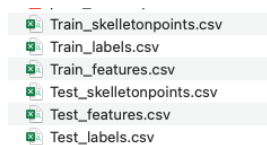
- Real Wi-Fi measurements were taken when people were doing specific poses.
 - Walk, sit down, run, stand up, bend...
- The features are the Wi-Fi measurements (the CSI, in this case).
- The labels (poses) were taken using cameras (Alphapose).

^aZhou, Y., Xu, C., Zhao, L., Zhu, A., Hu, F., & Li, Y. (2022). CSI-former: Pay more attention to pose estimation with WiFi. *Entropy*, 25(1), 20.



Dataset (II)

- Train features (`Train_features.csv`): CSI measurements in the training dataset.
 - 1000 samples $\times$ 270 (a flattened CSI matrix of $30 \times 3 \times 3$)
- Train labels (`Train_labels.csv`): actual pose in the training dataset.
 - 1000 samples of an integer between 1 and 5 (`{'wave', 'push', 'crouch', 'sitdown', 'bend'}`)
- Test features (`Test_features.csv`): CSI measurements in the test dataset.
 - 200 samples $\times$ 270 (a flattened CSI matrix of $30 \times 3 \times 3$)
- Test labels (`Test_labels.csv`): actual pose in the test dataset.
 - 200 samples of an integer between 1 and 5 (`{'wave', 'push', 'crouch', 'sitdown', 'bend'}`)



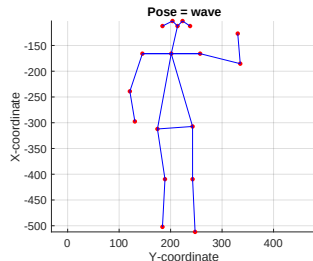
In this seminar, we will use a simplified version of a dataset that can be found at:
<https://github.com/NjtechCVLab/Wi-PoseDataset?tab=readme-ov-file>

Dataset (III)

In addition, SkellectionPoints are provided:

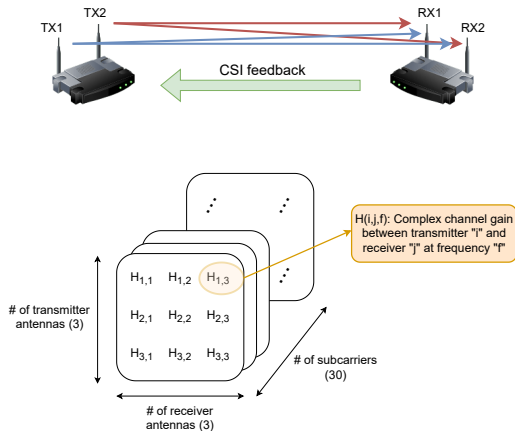
- Only for **visualization purposes**
- Each skeleton contains 54 points:
 - x coordinates (from 1 to 18)
 - y coordinates (from 19 to 36)
 - Confidence of each coordinate (from 37 to 56)
- Pose data are encoded using MPII^a

^a<https://www.mpi-inf.mpg.de/departments/computer-vision-and-machine-learning/software-and-datasets/mpii-human-pose-dataset>



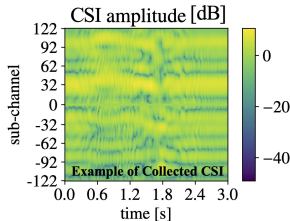
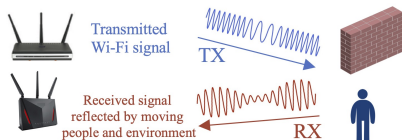
```
% Define connections between the 18 keypoints
connections_18 = [
    1, 2; % Nose - Neck
    2, 3; % Neck - Right Shoulder
    3, 4; % Right Shoulder - Right Elbow
    4, 5; % Right Elbow - Right Wrist
    2, 6; % Neck - Left Shoulder
    6, 7; % Left Shoulder - Left Elbow
    7, 8; % Left Elbow - Left Wrist
    2, 9; % Neck - Right Hip
    9, 10; % Right Hip - Right Knee
    10, 11; % Right Knee - Right Ankle
    2, 12; % Neck - Left Hip
    12, 13; % Left Hip - Left Knee
    13, 14; % Left Knee - Left Ankle
    1, 15; % Nose - Right Eye
    1, 16; % Nose - Left Eye
    15, 17; % Right Eye - Right Ear
    16, 18; % Left Eye - Left Ear
    3, 6; % Right Shoulder - Left Shoulder
    9, 12; % Right Hip - Left Hip
];
```

Channel State Information (I)



- Each element contains the channel gain for each **antenna pair** and **subcarrier**
- In the dataset, we have 5 CSI data packets of $30 \times 3 \times 3$ corresponding to 1 image (1 pose)
 - 30 are the number of subcarriers
 - 3×3 are the 3 Wi-Fi transmitter and receiver antennas

Channel State Information (II)



Wi-Fi sensing: detect movement, presence, and other characteristics of people and objects in a given environment thanks to CSI feedback, RSSI, or other measurements (e.g., packet arrival time).

